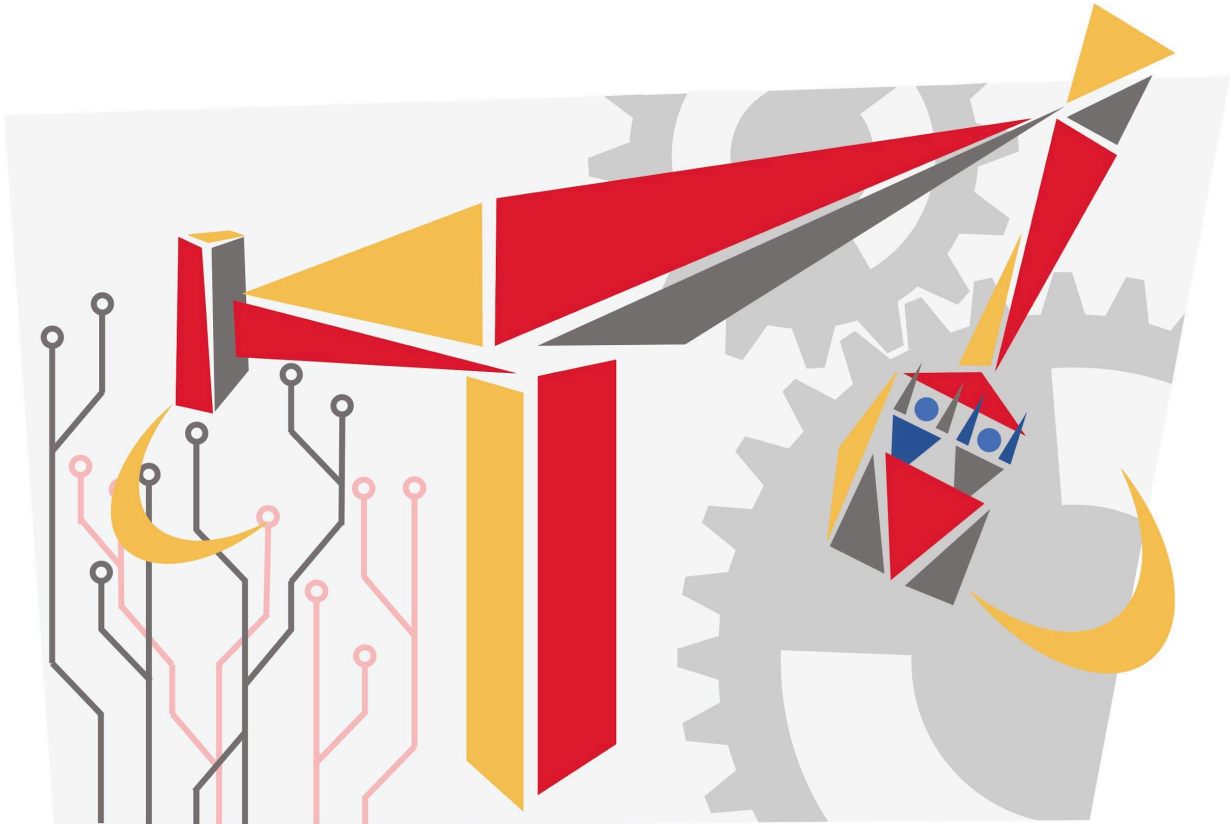


Ride Engineering Competition



**2026 Prompt
Thrills on the go:
Portable Ride Engineering Challenge**

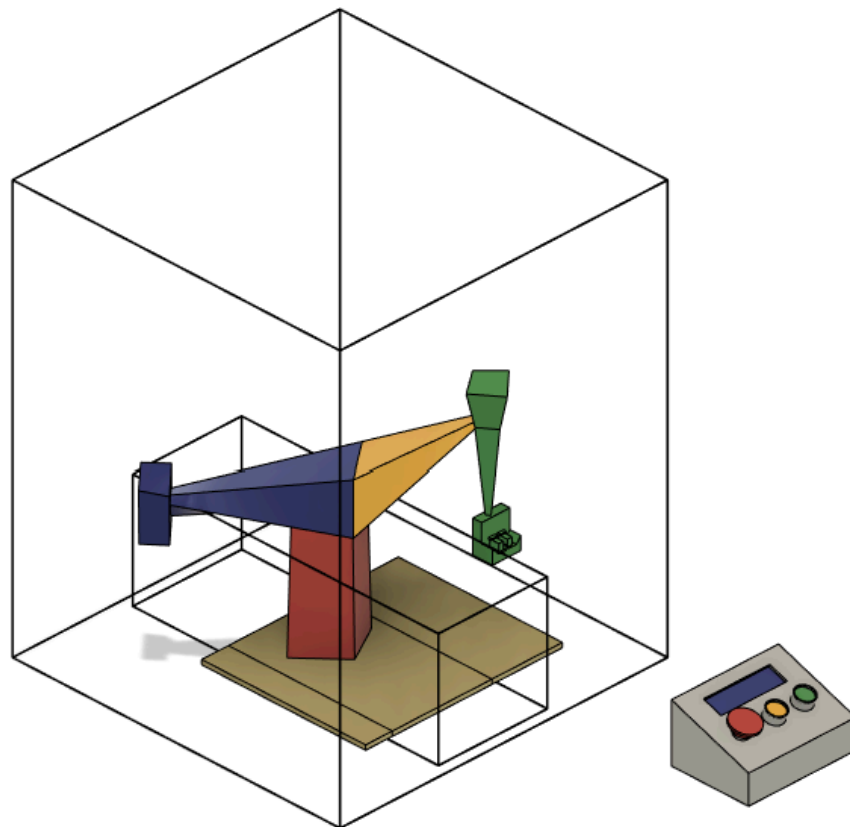
Intro

REC continues to travel to new locations and we need rides that are designed to do the same. We need products that pack a thrilling experience into an easily transportable size, engineered to transform between the two.

Overview

In this year's challenge, Teams will design, engineer, and manufacture an amusement ride that will operate inside an Operational Envelope. The ride will be disassembled into a Portable Envelope. Riders need to experience height, rotation, and acceleration in the ride experience. Each team will run their Ride for **6** hours. In the middle of the operation day, teams will disassemble their ride into the Portable Envelope, and reassemble it to resume operation.

Note on rider scale: This is not a model ride. The real riders of this attraction are candies as described in the Rule book. These riders are not stand-ins for human riders. These candy riders are the real riders of this real attraction. The accelerations and forces used to engineer the ride shall be based on the actual accelerations and forces experienced by the candy riders.



Design

Returning teams must not use the same exact ride concept used in the previous two years by members of the academic institutions they represent.

Ride designs may emulate existing rides, be an original design, or anywhere in between.

Note: The REC does not hold any Ride Design Intellectual Property and does not provide any Intellectual Property protection to ride designs introduced in the Ride Engineering Competition. All materials delivered to Ride Engineering Competition Inc. will be considered publicly disclosed.

Ride systems may make use of any Commercial-Off-The-Shelf (COTS) parts or custom manufactured parts.

COTS parts or kits that are designed for the intention of emulating an Amusement Ride may not be used in the specific amusement ride function they were designed for.

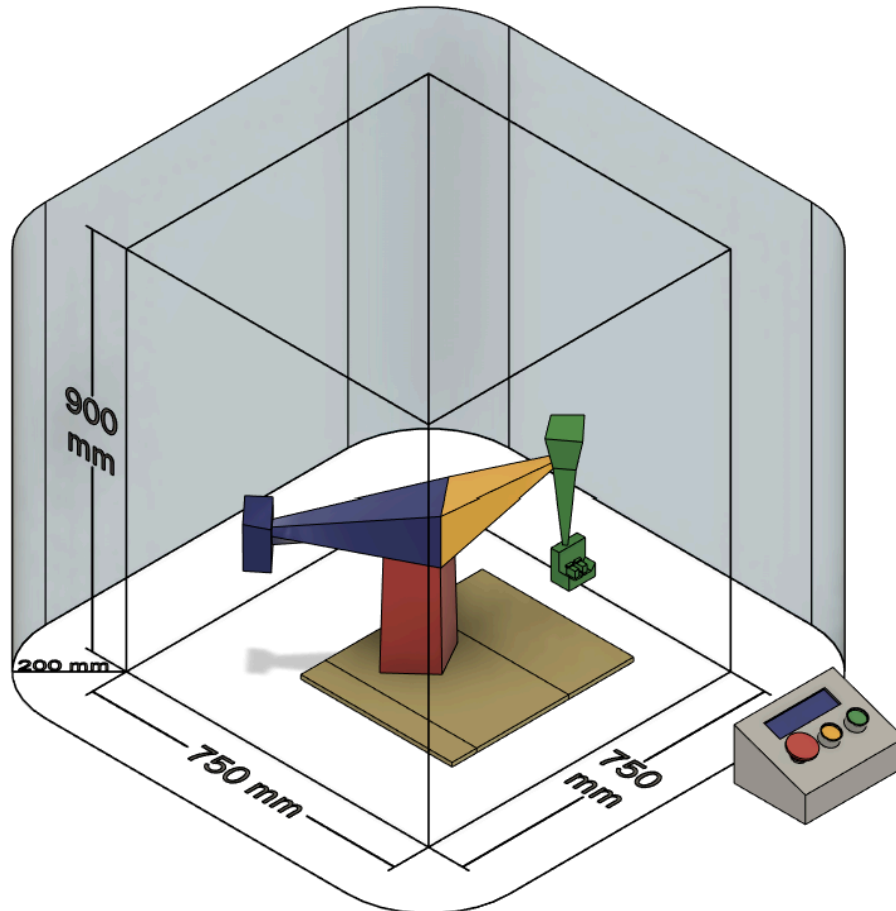
E.g. a K'nex brand Track ties and rails, may not be used as track ties and rails, respectively, within the Ride Design. These parts may still be used in the design in any other configuration or use case. For any needed clarification on this rule, please contact the REC.

All Rides must include an Emergency Stop function. The function shall be Red in color and be clearly labeled "Emergency Stop" or "E-Stop". Activating the function shall initiate a Ride Off Function Consistent with ASTM F2291 Section 11.3.8.

Surfaces that make contact with the Table must be free of burrs and sharp edges to not damage venue surfaces.

Operational Envelope

Rides will stand assembled within a 750 x 750 x 900 mm box. Portions of the Ride may extend up to 200mm outside the Envelope during operation. The base of the construction must be entirely contained within the 750 x 750 mm square base. Teams shall assume that there is no surface outside of the base.



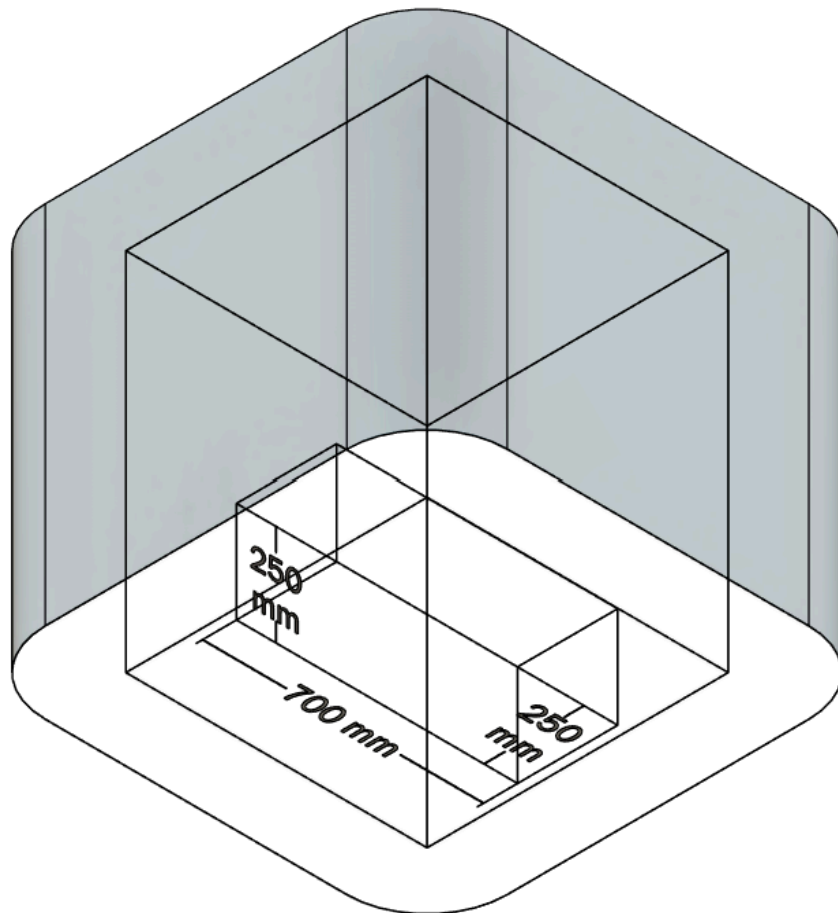
Operation Interfaces may be located outside the Operational Envelope, however any components that do not require physical operator access during the Ride's operation (such as power supplies and programmable controllers) must be located within the Envelope.

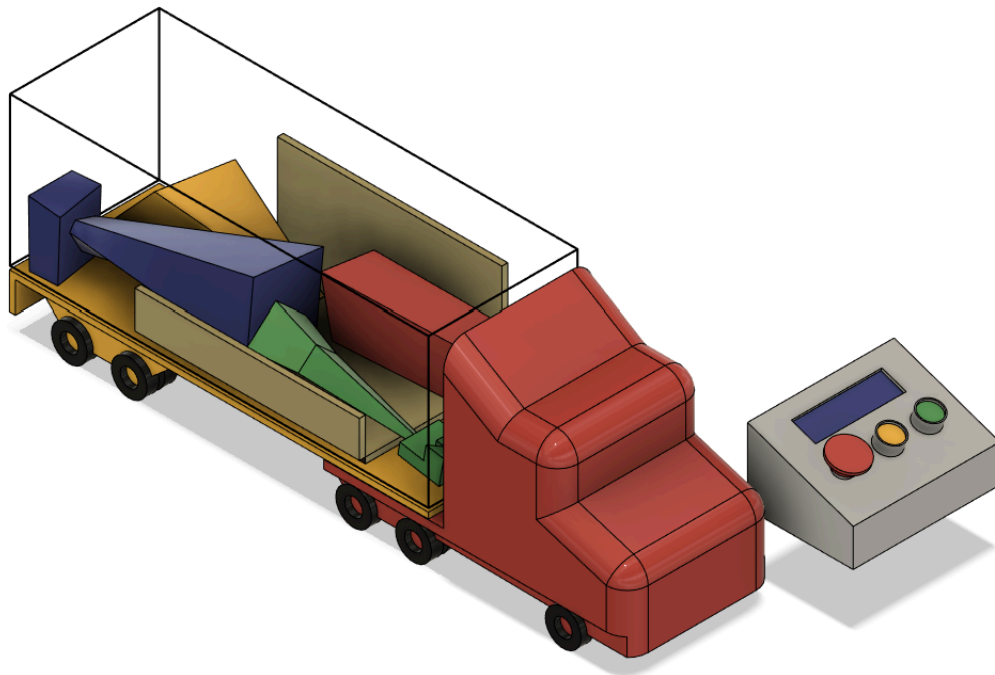
E.g. A series of Push Buttons may be placed outside the Operational Envelope, but a power supply or control board must be placed inside the Operational Envelope. If the board has a reset button that is only used when the ride is not operating, it must be inside the Operational Envelope.

Connection Cables to a power outlet may extend outside the Operational Envelope.

Portable Envelope

When disassembled, rides shall be contained within a 700 x 250 x 250 mm area, freestanding without any external support. The disassembled ride should be truly portable, and able to be placed on a flat 700 x 250 mm bed. All components of the ride including transportation devices (Side walls, Containers, Straps, etc.) shall be contained within this envelope. A Clearance Gauge shall be placed over the disassembled ride by REC Staff to ensure all components are contained within this envelope.





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Connection Cables to a power outlet may extend outside the Portable Envelope.

Rider Experience

Height

Each Rider shall be lifted 300mm above their initial loading height at least once during every ride cycle.

Orientation

Riders shall change orientation by at least 45 degrees in any direction during the ride cycle. Riders can end a ride cycle in a different orientation than at the start of the ride cycle.

Acceleration

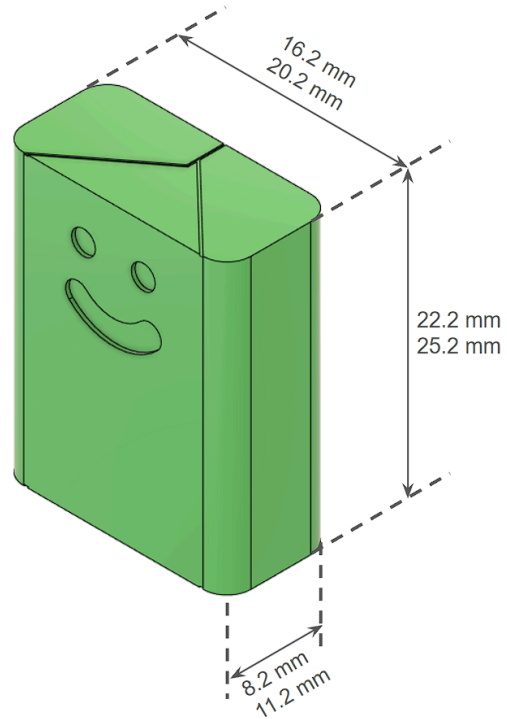
Riders shall experience at least 2G (19.62 m/s^2) of absolute acceleration force in any direction at least once during every ride cycle. This includes acceleration from gravity. This is a real acceleration that can be measured by an accelerometer, not a theoretical scaled acceleration.

Riders

The riders are approximately 1:50 the scale of an Adult Male Human rider. Specifically, the riders shall be a rectangular prism represented in size and weight as a Jolly Rancher Chews® candy. These riders are not human, and should not be assumed to have human qualities unless determined by your team. The specific rider qualities determined by a team may vary between teams. Please take care to communicate your determinations in your reports.

The Riders have the following approximate dimensions:

Height	22.2 - 25.2 mm
Depth	8.2 - 11.2 mm
Width	16.2 - 20.2 mm
Weight	4 - 5 grams



An STL file of the 2026 Rider is available from the REC Website and can be 3D Printed. Printing the file with PLA filament at 60-70% infill will create a rider of a nominal size and weight.

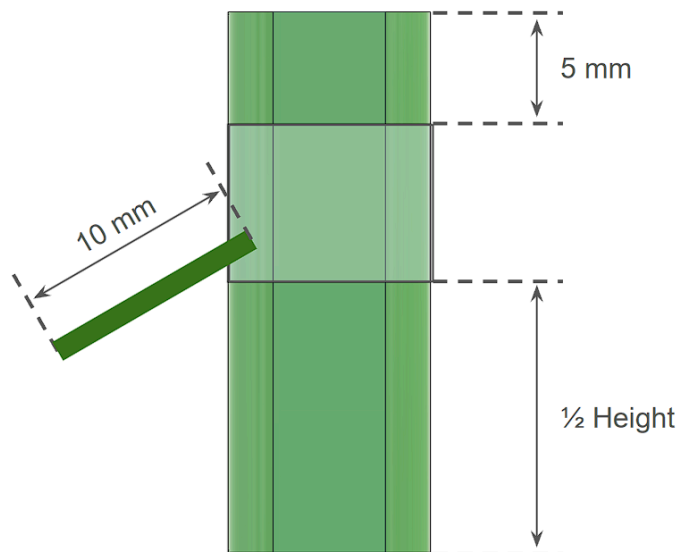
At the final event, candy Riders will be provided to all teams and must be used. No 3d printed riders are allowed during the competition day.

Arm Reach:

Teams shall include in their risk assessment that riders have 'arms' that are 10 mm in length and can protrude from the sides of the rider in any direction, starting at a Shoulder anywhere within a rectangular area on the sides of the rider. The Shoulder area starts at half the rider's height, is 5 mm from the top and meets the front and back as described in the following figure.

These 'arms' will not be represented in any physical way at the competition but shall be accounted for in the design.

Rider Arm Shoulder Area



Operation Day

Teams will have at least 30 minutes total to unseal their Ride and prepare it for operation. Rides must be verified to fit within the Portable Envelope Clearance Gauge by an REC Staff member before assembly can begin.

The Operation Day will begin at a set moment for all teams.

Any work on the Ride after the Start of the Operation Day will be considered downtime.

Rides will operate during a period of 6 hours.

In the middle of the Operation Day at a prescheduled time, Teams will be required to halt operation and disassemble their ride back into the Portable Envelope. Once a Ride has been verified to fit within the Portable Envelope Clearance Gauge by an REC Staff member, that team may reassemble their ride and attempt to return to operation.

Rides will operate primarily in Continuous Automatic Operation, where operator intervention (e.g. pressing a dispatch button) is **not required** between Operation Cycles.

Operation Cycle: An Operation Cycle consists of the ride program, followed by a pause of approximately 0.5 seconds per rider to simulate loading and unloading.

There will be a minimum of 8 riders allowed on the ride during a single cycle. There is no maximum riders per cycle limit.

Rides will have a throughput between 32 and 64 riders per minute when in continuous automatic operation. Throughput = (# of Riders per Cycle) / (Operation Cycle duration).

Example:

A ride has 20 riders and a ride program lasting 20 seconds. The Operational Cycle duration is 20 seconds + (20 riders x 0.5 seconds/rider) = 30 seconds = 0.5 minutes. The throughput of the ride is 20 riders / 0.5 minutes = 40 riders per minute.

Riders will stay on the ride for one hour as the ride operates in Continuous Automatic Operation. Every hour, on the hour, Teams shall safely stop the Ride. The riders must be then removed from the ride and new riders shall be placed on the ride.

Note on rider scale: This is not a model ride. The riders of this attraction are candies as described in the Rule book. These riders are not stand-ins for human riders. These candy riders are the real riders of this real attraction. The accelerations and forces used to engineer the ride shall be based on the actual accelerations and forces experienced by the candy riders.

Facilities input

Each Team will have access to:

A table for their ride and any tools or equipment they need to service their ride.

1 Standard 120VAC Outlet for the Ride.

1 Standard 120VAC Outlet for additional Tools or Equipment.

Teams may not draw more than 10 Amps total.

See the Safety Chapter of the 2026 REC Rule Book for more details about allowed tools and PPE at the event.

Travel

The competition event will be held in person. Teams are responsible for transporting their ride.

The REC Planning Committee has determined that the value to competitors, Judges, and the competition as a whole gained by having rides and participants in the same physical space is worth the travel logistics burden on teams.

Teams should consider the transportation of their ride in their design to ensure that their ride can be made functional on competition day. The Portable Envelope is designed to fit within most large suitcases. On the day of competition, teams will have less than an hour in the event venue before operation begins and should be prepared to have their ride operational within that time.

Rides will be sealed in a bag per the Bag & Tag section of the Rule book when they arrive at the Event. Competition officials will conduct an inspection before instructing teams to unseal their ride. Rides must be verified to fit within the Portable Envelope Clearance Gauge before assembly can begin.

For specific questions about transportation or if a team is completely unable to bring representatives or their ride to the event, please contact the planning committee and we will assist you on a case by case basis.

Good Luck Ride Engineers!